



TeamRob Subproject 3: Mixed Human Robot Teams

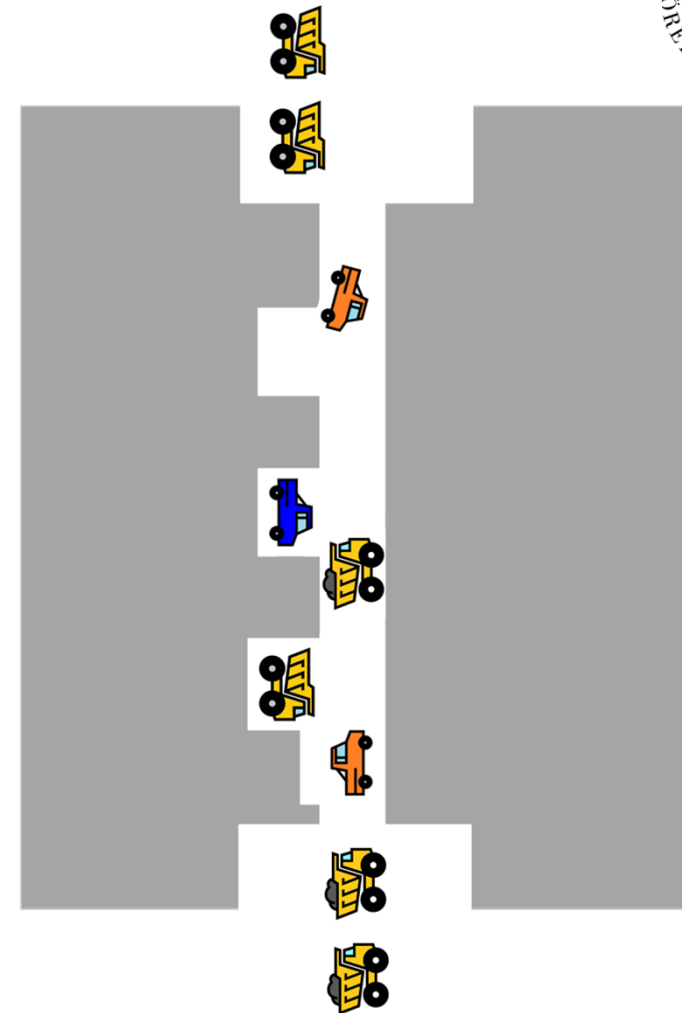
Fabien Lagriffoul, (Ahmed Nouman), Franziska Klügl

Mixed Human-Robot Teams – Level of Decision Making

	How Coordinate? Challenges?
Task Planning	Coordination is build-into the planning process, sequencing of steps between agents, uncertainty on human actions, information, replanning
Motion Planning	Post-Planning by analysing future trajectories, sort access to conflict areas, uncertainties of human's positions, traffic control
Control	Emergency Stop, Reliable detection, Constraints on design, movement...

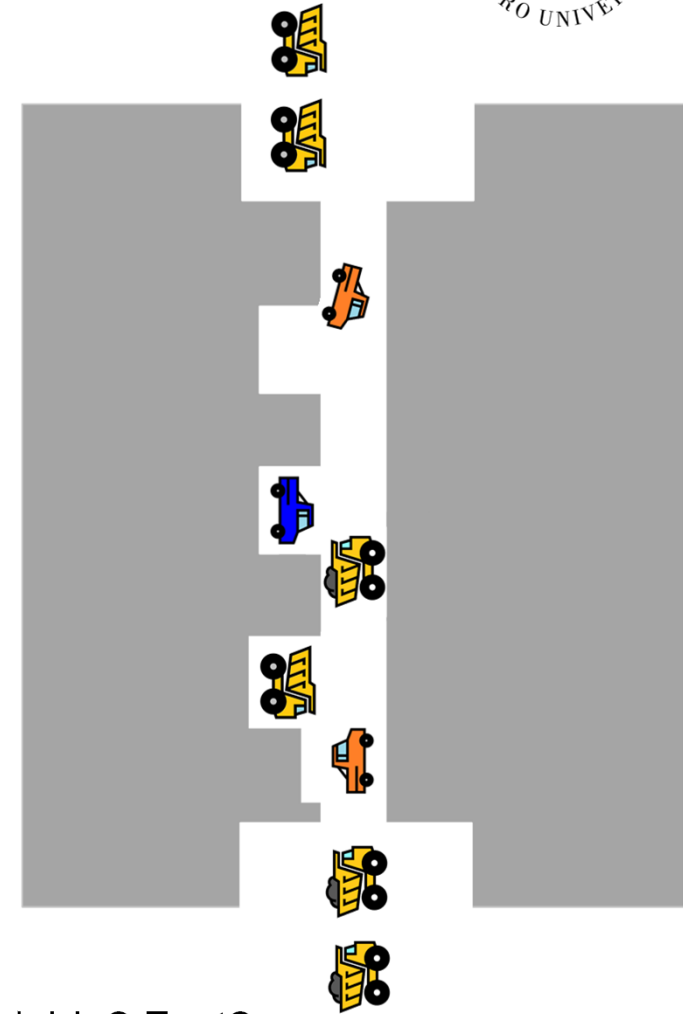
Simple (?) Use Case: Traffic Management for Ramps

- Two opposing flows of vehicles traverse a ramp as a bottleneck - one lane, no traffic can pass nor overtake.
- Some vehicles going upwards cannot stop
- Stoppable vehicles can use passing places to give way
- Queues at top and bottom entrances
- Individual speed, ramp size?
- ***Must be solved on the planning level***
Dynamic situation → fast and scalable



Modelling the Ramp Problem for A*

- All vehicles are agents with actions that need to be coordinated → Multi-Agent Planning
- Continuous state: agents positions and current speed
→ no discretization of space and time
- “Mini-Simulator” to check for collisions (second layer of filter) and to determine next state → “Event-based state generation”
 - Activity has terminated (Vehicle as arrived at local destination)
 - Vehicle passed evasion place
- Individual agent actions with logical and geometric preconditions
first layer of filter
- Start with deterministic, fully predictable and compliant agents → scalable? Fast?



Fabien's Demo / Video

What to do next?

- Further testing with realistic ramp setting
- Prepare publication
- Integrating humans
 - Unclear speed
 - Do they follow instructions? How do we give instructions?
→ When trigger re-planning? May this lead to intractable start situations?
- Excavator Fleet Scenario (CivIng Exam Work) ?